Marten Weijer

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About

I am a master's student in Data Science at Minnesota State University, Mankato, advised by Prof. Rushit Dave. I'm drawn to questions about how AI is changing software development, something I saw first-hand during my years as a software engineer at Booking.com. My interests include AI for software engineering, empirical software engineering, and software analytics. I am applying to PhD programs for Fall 2027.

Research

Beyond Merge Rates: Measuring Abandoned Pull Requests from AI Coding Agents
M. Weijer
Under review at a software engineering conference

Does Pseudo-Labeling Help Rare CWE Classes? An Empirical Study with CodeBERT
M. Weijer, S. S. Abdullah, S. Alkhushayni
Preprint, 2026

Interpretable Machine Learning for Municipal Residential Gas Consumption: Integrating Climate, Building Stock, and Socioeconomic Drivers
M. Weijer, R. Dave, M. Bhavsar
Sustainability (MDPI), Special Issue on Artificial Intelligence, Image Processing, and Data-Driven Solutions for Sustainable Development, 18(20), 10407, 2026
doi:10.3390/su182010407

The Missing Middle: A Systematic Review of Machine Learning for Municipal-Level Residential Energy Forecasting
M. Weijer, R. Dave, M. Bhavsar
2026 IEEE International Conference on Emerging Trends in Networks and Computer Communications (ETNCC), Windhoek, Namibia, 2026
Accepted and presented; proceedings forthcoming

Education

Minnesota State University — Mankato, MN
Master of Science in Data Science — *Jan. 2026 – May 2027*

Amsterdam University of Applied Sciences — Amsterdam, The Netherlands
Bachelor of Science in Computer Science — *Aug. 2013 – May 2017*

Experience

Graduate Teaching Assistant — Aug. 2026 – Present
Minnesota State University – CIS 483: Web Applications and User Interface Design — *Mankato, MN*

- Grade assignments, hold lab office hours, and give code-review feedback on student web applications

Senior Software Engineer — Feb. 2021 – Jan. 2026
Booking.com — *Amsterdam, The Netherlands*

- Designed and maintained large-scale data systems, with a focus on testing, code review, CI and reliable delivery across teams
- Built and operated backend services and event-driven pipelines in Java, Spring and Kafka, deployed on Kubernetes
- Led design discussions and code reviews, and mentored engineers on testing practices and maintainable code
- Took part in on-call rotations, debugging production incidents and following up with root-cause analysis

Software Engineer — May 2017 – Feb. 2021
Capgemini B.V. — *Amsterdam, The Netherlands*

- Developed Python and SQL pipelines and automated test suites, working in version-controlled, CI-driven development workflows
- Built and maintained backend applications and REST APIs for enterprise clients across several projects
- Worked in Scrum teams, turning client requirements into technical designs and shipping features in regular sprints

Projects

Sneakerjagers.com | *PHP, Node.js, PostgreSQL, Elasticsearch, Kafka* 2015 – Present

- Sole developer of a sneaker release calendar and price-comparison platform with around one million unique visitors per month, available in Dutch, German and English
- Built the full stack: frontend, backend, database design, and native iOS and Android apps
- Ingested and indexed product and price data from many retailers, using Kafka for data pipelines and Elasticsearch for search
- Owned testing, deployment and operations end to end, as a side project alongside full-time engineering roles

MinneMUDAC 2026 Data Science Hackathon | *Python, Jupyter, pandas, scikit-learn, XGBoost* Sep. – Oct. 2026

- Won the award for Best Prediction Model
- Graduate-division team of four analyzing food shelf visit and demographic data from The Food Group to address food insecurity in Minnesota
- Built the team's prediction model in a Jupyter notebook using machine learning models including XGBoost, used to identify the counties where The Food Group should focus its efforts

Certifications

Kafka: Confluent Certified Developer for Apache Kafka (CCDAK) 2024
SAFe: Certified SAFe Practitioner 2024
Spring: Spring Certified Professional for Java Developers 2024
Kubernetes: Certified Kubernetes Application Developer (CKAD) 2023
Scrum: Professional Scrum Master I 2023
Java: Oracle Certified Java SE Developer 2022